The Model Knows Better Than It Answers: Gemma 3 Composes a Spatial Map Its Answers Never Read

Cristhian Gómez Neira

Abstract

When people read "A is 3 units north of B; C is 2 units east of A", they can answer questions the text never answers, as if looking at a picture. We ask whether a language model builds such a picture, and whether its answers use it. On text-only layouts, Gemma 3 12B builds a map: linear probes read each object's position from its activations ($R^2 = 0.74$), including positions that require combining sentences. But the answers do not use this map. Erasing it everywhere costs no more accuracy than erasing random directions. Instead, the answers depend on a small 2-D code over the sentence that states each position. Swapping it moves 72% of the affected answers (83% in Gemma 3 27B; random: 0%), and it acts as a switch, not a coordinate. The answers also use the chain of relations only partly, so answering from the model's own unused map beats the model on chain questions (0.80 vs 0.73). The main tests were preregistered.

1. Introduction

Does a language model solve a spatial puzzle by consulting something like a map, or by chaining the relations it was told? Written reasoning cannot settle this, because it contains only words. The question is about the format of the model's internal computation.

Geometry can be read out of language models. Probes recover the coordinates of real places (Gurnee & Tegmark, 2023), circular codes for days and months (Engels et al., 2024), and spatial relations in described scenes (Tehenan et al., 2025). Whether such representations are used is less clear. An et al. (2026) find spatial information that "can causally influence behavior" but is "weakly integrated into final predictions". Probing alone cannot answer the question: a probe shows that information is present, not that the model relies on it (Hewitt & Liang, 2019).

We test use directly. If a model answers from a map, then editing one object's position in that map should move every spatial answer consistently, and removing the map should hurt. We find the opposite. The map is real and accurate, but the answers come from a different route. Our contributions:

- A text-only layout task, with a behaviour test that selects a model able to solve it without written reasoning.
- Evidence that Gemma 3 12B composes object positions across sentences, including distances that rule out simple copying of the stated offsets.
- Evidence that the answers do not use this map: copying it, steering it and erasing it all leave the answers unchanged, also on harder 4–5-object scenes.
- A small 2-D code at each stated sentence that does control the answers, confirmed on fresh data and replicated in Gemma 3 27B. It behaves like a switch, not a coordinate.

- A mechanism: the question reads each object's own sentence, never the map, and uses the chain only partly. Answering from the unused map would be more accurate on chain questions, but pointing the model's attention at the map does not help.

2. Setup

Scenes and questions. Each scene places three named objects on an integer grid, using relations such as "Cy is 2 units east of Ana". One object is the root. An object placed relative to an object that is itself placed relative to the root is two steps away; its position follows only by combining two sentences. After the description, the prompt lists the objects ("The objects are: Ana, Bo, Cy."). Questions ask which object is farthest from or closest to another, the compass direction from one object to another, and which of two objects is further north or further east. We score each allowed answer by its log-probability, without generation. Correct answers win by at least 1 unit (10° for direction).

Model. We screened eight open models (Qwen3 1.7B–14B, Llama 3.2 3B, Gemma 3 4B–27B) with a preregistered behaviour test. The smallest model that solved 3-object scenes reliably without written reasoning was **Gemma 3 12B** (accuracy 0.76 [0.74, 0.77]; Fig. 1). Gemma 3 27B served as a replication. One question type ("Is Y closer to X than Z?") was at chance for every model and was excluded.

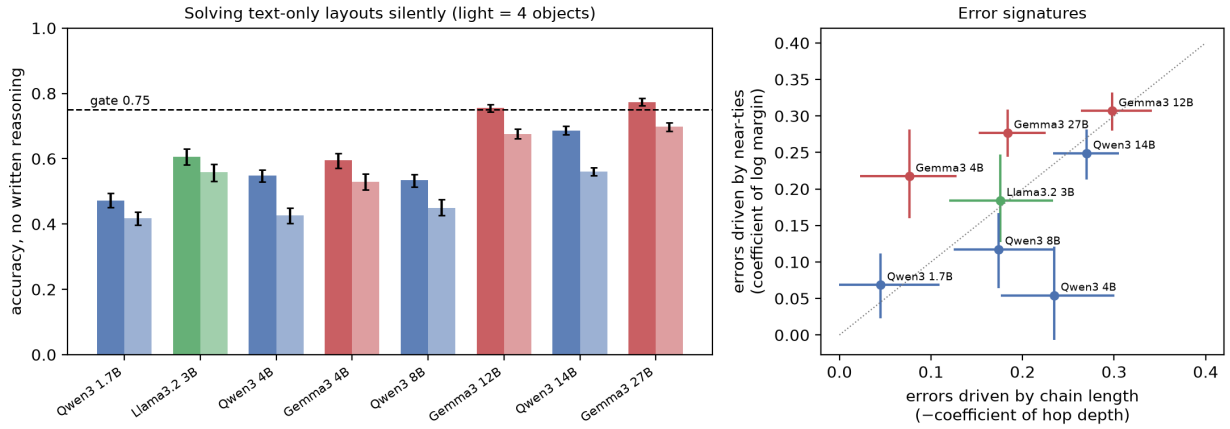


Figure 1: Left: accuracy without written reasoning (dashed line: the preregistered threshold). Qwen models put little probability on one-word answers, so their accuracies are lower bounds. Right: errors grow with chain depth and with how close the competing answers are.

Methods. We use four kinds of tools.

- *Probes:* ridge regressions that read each object's (x, y) from the residual stream at its name in the object list, with shuffled-label, layer-0 and random-model controls.
- *Activation patching:* run the model on one scene but copy in hidden states from a scene where one object sits elsewhere.
- *Distributed alignment search* (DAS; Geiger et al., 2024): learn a 2-D subspace whose swap moves the answers.
- *Erasure:* iterative nullspace projection (Ravfogel et al., 2020) removes the directions a probe reads.

Unless stated otherwise, causal metrics compare the edited answer with the answer the model itself gives on the scene being imitated.

Preregistration. We committed hypotheses, metrics and decision rules to version control before collecting each dataset. Section 6 lists every test, its verdict, and which analyses were exploratory.

3. Results

3.1 The model builds a map

A linear probe at the object list reads each object's position. Accuracy is near zero up to layer 11, rises sharply at layers 12–13, and peaks at **layer 21 ($\mathbf{R^2 = 0.74}$ [0.72, 0.75])**. All controls score 0.00 (Fig. 2a).

Objects two steps from the root are read as well as one-step objects (0.74 vs 0.76). A baseline that knows only each object's own stated offset reaches 0.22 on them, so the model combines the sentences. It also computes something beyond a sum of offsets. A probe for distance to the root reaches $\mathbf{R^2 = 0.62}$ **[0.56, 0.67]** on two-step objects, while the best additive baseline built from both stated offsets scores −0.05. The same coordinates can be read at the object's name inside a question and at the end of its own sentence.

3.2 The answers do not use the map

We tried to make the answers follow the map in four ways. None worked.

- **Copying.** Copying an object's list representation from a scene where it sits elsewhere changes no answer: 0.00 [0.00, 0.00] on 390 pairs. Copying its name inside the question also gives 0.00.
- **Steering.** Moving the list representation along the probe's (x, y) directions, over ±5 units, flips an answer in only 2% of the 268 sweeps where geometry says an answer must flip.
- **A learned 2-D subspace at the list.** DAS at the object list reaches only 0.07 [0.05, 0.09].
- **Erasure.** We removed the probe-readable coordinates at every token, at layer 14, 21 or 28 alone, or at all layers 12–30. After erasure the probe's R^2 is at most 0.09 (down from up to 0.69). On 300 new scenes, erasing the map hurts questions about two-step objects no more than erasing random directions of the same rank. The excess drop ranges from −0.027 to −0.003, and no confidence interval rises above 0.001 (Fig. 3).

The last test could fail simply because three-object scenes are easy to answer by chaining. We therefore repeated the erasure on 4–5-object scenes, where questions can need up to four chained relations. The model still had room to fail: it answered deep questions with accuracy 0.65, against about 0.36 for chance. Erasing the map again did not hurt deep questions more than random erasure. The largest effect, 0.023 [0.004, 0.041] at layer 21, is a fifth of the preregistered threshold of 0.10.

Where does the information go instead? Copying an object's whole *sentence* moves almost all answers at layer 6 (0.94). This effect fades between layers 16 and 32, and over the same layers the question tokens take it over (0.78 by layer 32; Fig. 2b). The information flows from the stated sentences into the question and bypasses the object list.

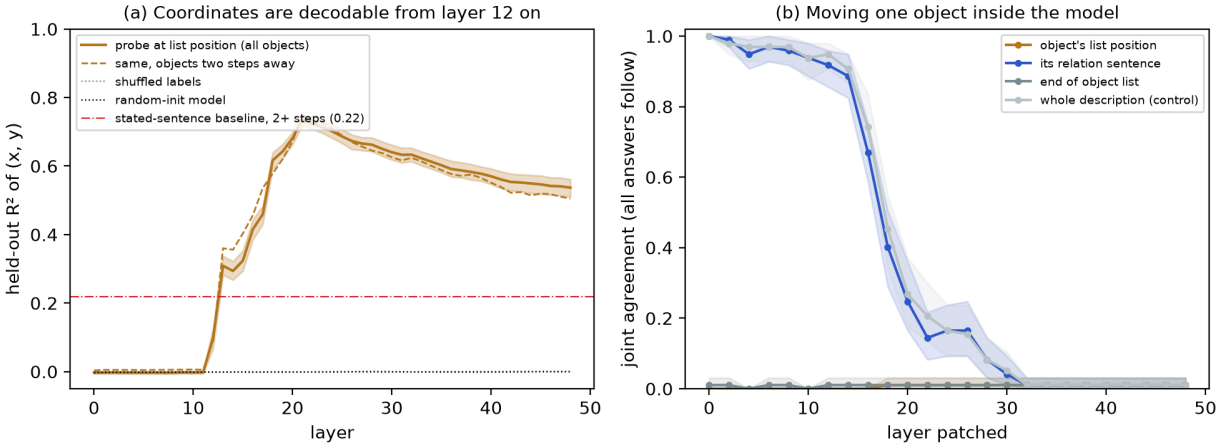


Figure 2: (a) Probe R^2 for (x, y) at the object list, by layer, with controls and the stated-sentence baseline. (b) Share of pairs in which every answer follows the copied site, by site and layer (100 pairs).

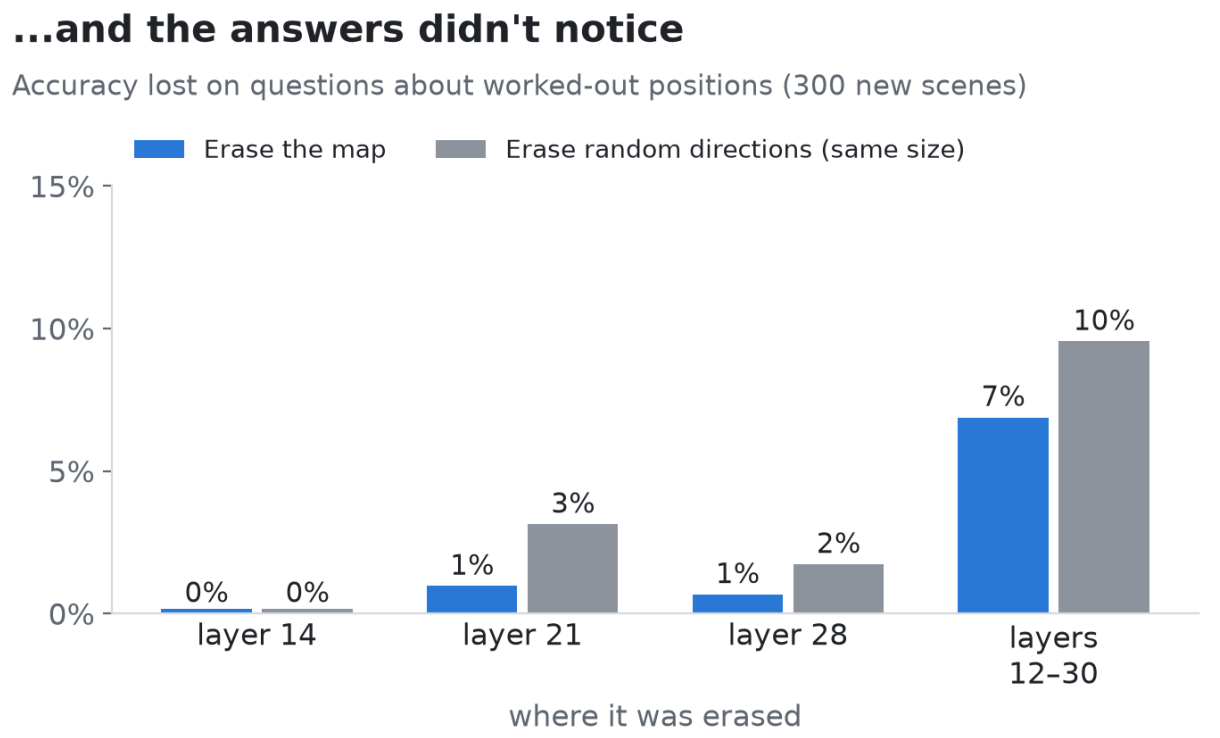


Figure 3: Accuracy lost on questions about two-step objects when the map is erased, compared with erasing random directions of the same size (Gemma 3 12B, 300 new scenes).

3.3 What the answers use instead: a 2-D switch at each sentence

Since the sentence carries the information, we searched for a compact code there. In an exploratory run, a 2-D subspace at layer 14 of the moved object's sentence moved 0.81 of the affected answers. We then fixed the site, layer and decision rule in advance and reran on fresh scenes, adding Gemma 3 27B at the matching relative depth.

	Gemma 3 12B (layer 14)	Gemma 3 27B (layer 18)
Learned 2-D code, all questions	**0.72 [0.68, 0.75]**	**0.83 [0.80, 0.85]**
Random 2-D code	0.00	0.00
Further north / further east	0.94 / 0.90	0.94 / 0.84
Direction	0.60	0.81
Closest / farthest	0.38 / 0.33	0.43 / 0.38

Both runs pass the preregistered rule. The code works best for one-coordinate questions, and worst for closest and farthest, which depend on every object at once. Three checks suggest it is a genuine part of the computation rather than an artefact of the method (Makelov et al., 2024).

- *Stability.* Two independent trainings on disjoint data find overlapping planes (principal angles 47° and 56°; random planes never exceed a cosine of 0.07).
- *Competence.* The edit succeeds more often where the model itself answers correctly (0.75 vs 0.48 in 12B).
- *Write-in.* Setting the code from the stated offset, instead of copying it from another run, still moves 0.27 of the answers, against 0.00 for a random subspace.

It is a switch, not a dial. If the two numbers were coordinates, setting them halfway between two layouts should move the object halfway. We tested this on questions whose answer at the halfway position differs from both stated layouts. The model rarely gives that in-between answer: **0.085 in 12B and 0.055 in 27B, below the 0.141 expected by chance**. Instead it snaps to one of the two stated layouts (Fig. 4b). Consistent with this, the code does not change linearly with the stated distance: a linear fit explains it no better than for random directions (R^2 0.31 vs a 95th percentile of 0.35).

3.4 Why the map is not used

We now ask what the question reads, using chains R → M → C, where C is placed relative to M and M relative to the root R.

The question reads each object's own sentence, never the list. We blocked the question tokens from attending to one part of the prompt. Blocking the object list, where the map lives, changes accuracy by at most 0.018. Blocking C's own sentence costs questions about C and M **0.35**, and blocking M's sentence costs questions about M and R **0.41**. Nearly all of these effects come from layers 16–31.

But it does not re-read the chain. Questions about C and R need M's offset. Yet blocking M's sentence costs them only **0.027 [0.017, 0.039]**. This rejects our preregistered hypothesis that the question re-reads every sentence in the chain.

The answers use the chain only partly. If M's offset reaches the question, it must come through C's sentence. So we copied C's sentence, at every layer, from a scene where M (and therefore C) had

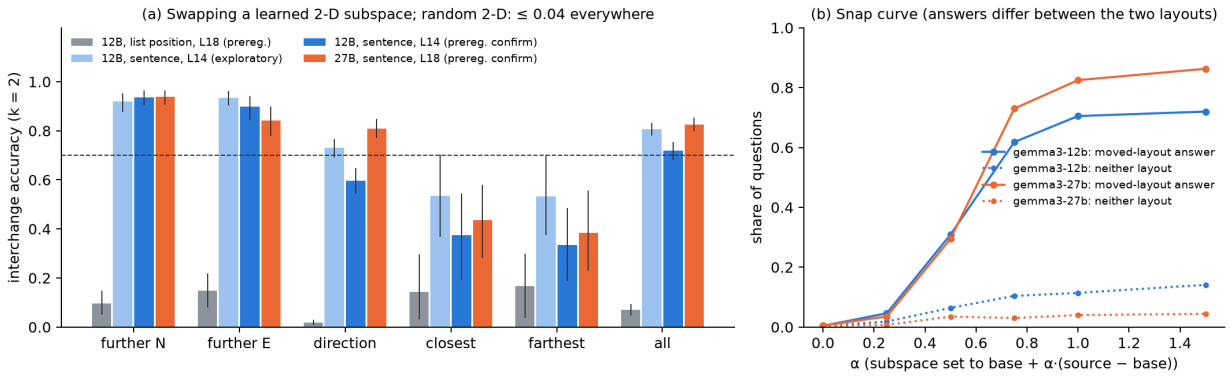


Figure 4: (a) Share of affected answers moved by a learned 2-D subspace, by question type. Grey: the object list. Blue and orange: the moved object's sentence, exploratory and preregistered runs. (b) As the subspace is set from the original value ($\alpha = 0$) to the moved value ($\alpha = 1$), answers switch abruptly from one stated layout to the other.

moved. This moved only **0.13 [0.09, 0.17]** of the answers. An exploratory comparison shows why. We built pairs in which C ends up at the same new position in two ways: either C's own sentence changes, or only M's sentence changes. The geometry is identical, so a model that fully combines the chain should update its answers equally often. It does not. On questions about C and R, the model updates correctly in **0.39** of cases when C's own sentence changes, but in only **0.21** when only M's sentence changes. In most of the second kind of case (72%), it does not change its answer at all. The answers lean on each object's own sentence and give the parent's offset about half its weight.

3.5 The map would have answered better

The map combines the chain correctly (§3.1), while the answers do not. So answering from the map should beat the model, but only on questions that need the chain. We decoded every object's position with the layer-21 probe, answered each question geometrically from the decoded positions, and compared these answers with the model's own on 300 new chains (5,350 questions).

The map wins exactly where expected (Fig. 5). On questions about C and R it scores **0.80 vs 0.73 for the model (difference +0.069 [+0.042, +0.094])**. On questions that need no chaining, the two tie (0.91 vs 0.89 and 0.90 vs 0.90). Layer 24 shows the same pattern (+0.056). The advantage depends on how well the probe reads the map: it shrinks at layer 28 (+0.010) and reverses at layer 18, where the probe is weakest.

Pointing attention at the map does not help. If the map were usable as it is, pushing the question to attend to it should improve chain answers. We added a bias of +1 to +8 to the attention from the question to the object names in the list (layers 16–31). A control applied the same bias to the token just before each name. No bias improved chain questions beyond the control. Strong biases hurt every question type. The one effect specific to the map was harmful: it lowered accuracy on questions about C and M by 0.049 (b = 2) and 0.136 (b = 4) beyond the control. Attending to the map distracts the question from the sentence it normally reads, without giving it a better answer.

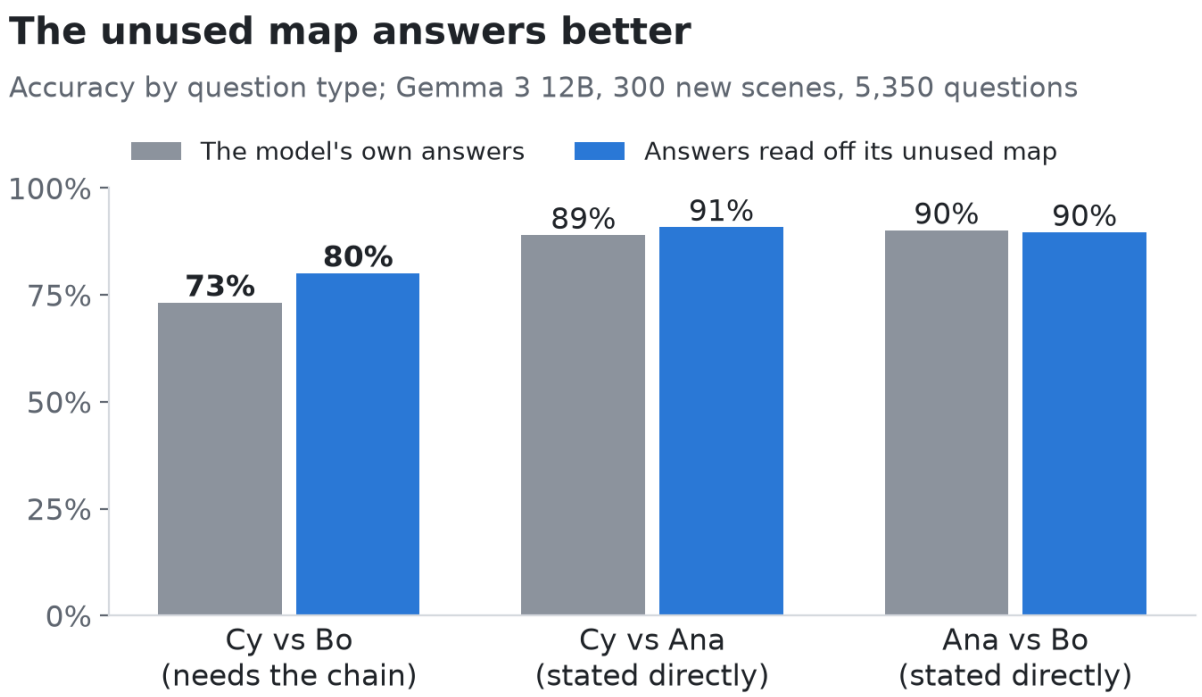


Figure 5: Accuracy of the model's own answers and of answers read off its unused map (layer 21), by question type.

4. Discussion

Summary. Gemma 3 builds an accurate map of the scene, combining sentences and even computing distances. Its answers take a different route. Each sentence is condensed into a compact, switch-like code, and the question reads the codes of the objects it names in layers 16–31. This route uses the chain of relations only partly. The map, built at the object list, sits off that route. It can be erased without cost, and it would answer chain questions more accurately than the model does. This is a spatial instance of a gap reported for other kinds of world state: a faithful internal state paired with an unfaithful read-out (Feng, Russell & Steinhardt, 2024).

Decodability is not use. Our probes pass every control we know of, including a nonlinear target, yet they sit off the causal path. Claims about "world models" in language models need an intervention on the decoded representation, and ideally an erasure with a random control (cf. Hewitt & Liang, 2019; Vafa et al., 2024). The contrast with Tehenan et al. (2025) fits this view. They steer answers along *relation* directions, and relation codes do steer answers in our model too; coordinate codes do not. The same point runs through the old debate on mental imagery, which was about whether depictive images do work in reasoning, not whether they exist (Kosslyn et al., 1978; Pylyshyn, 2002).

Why build a map at all? Our tests rule out one explanation: the map is not a backup that the model falls back on, at least not in its linear form. Erasing it everywhere costs nothing, even when chains are long. Two explanations remain. The map may be a by-product of how the model processes text in general. Or it may serve uses we did not test, such as describing a scene. Our question types can all be answered by chaining; questions that are hard to answer without a layout,

such as collinearity or containment, would test the second explanation. Since attention alone cannot connect the map to the answers, a natural next step is to ask whether light fine-tuning can.

5. Limitations

- **Scope.** Two models of one family, small scenes, five question types, fixed templates and integer grids. Other model families failed the behaviour test.
- **Linear erasure.** We removed only the linear subspace the list probe reads. Nonlinear or differently placed information may survive, so "not used" means not used in that subspace. The erasure, attention and map-answer tests were run on 12B only.
- **Probe-based answers.** The map's answers come from a probe we trained on the true positions. They show that the information is present and easy to read, not that the model could use it without extra computation. The advantage also depends on the layer.
- **Subspace methods.** The write-in recovers only 37% of the copy edit, so a linear reading of the stated offset captures only part of the code.
- **Exploratory analyses.** The distance probe, the 0.39 vs 0.21 comparison and the stability, failure and competence analyses were not preregistered.
- **Small classes.** Closest and farthest questions have wide intervals.

6. Preregistration

All rules are in `docs/preregistration*.md`, committed before the corresponding data.

Test	Prediction	Result	Verdict
Behaviour gate	a model solves the task without reasoning	Gemma 3 12B on 3 objects	met
Probes (P2)	coordinates readable, controls fail	R^2 0.74	met
Patching at the list (P3)	copying the map moves answers	0.00	failed
Probe steering (P4)	answers follow the probe directions	2% of sweeps	failed
DAS at the list (P4)	2-D subspace controls answers (≥ 0.7)	0.07	failed
DAS at the sentence (P4c)	≥ 0.70, random ≤ 0.05 (12B, 27B)	0.72, 0.83	confirmed
Halfway test	in-between answers below chance $\rightarrow$ switch	0.085, 0.055 vs 0.141	switch
Chain transfer, one layer (E3)	C's sentence carries its position	≤ 0.07	not carried
Natural encoding (P5a)	code is linear in the offset	R^2 0.31 vs null 0.35	no linear encoding
Write-in (P5b)	$\geq 50\%$ of the copy edit	37%	partial
Erasure, 3 objects (P5c)	erasing hurts beyond random	≤ 0	not used

Test	Prediction	Result	Verdict
Erasure, 4–5 objects (P6)	erasing hurts deep questions	≤ 0.023	still not used
Attention knockout (P7)	question re-reads the chain	0.027	rejected; list not read
Multi-layer transfer (P8)	C's sentence carries its position	0.13	partial
Map answers (P9)	map beats model on chain questions	0.80 vs 0.73	map better
Attention boost (P10)	attending to the map helps	no gain	no benefit

Deviations. A code review changed which model the unchanged selection rule picked, from Gemma 3 27B to 12B. One chain experiment (E2) also used a stricter question set defined after the data. In the attention-knockout test, a classification bug dropped direction questions about C and M before scoring. The verdicts rest on the other classes, which were unaffected. Each deviation is dated in `results/RESULTS.md`.

Code and data

Code, preregistrations (with their commit history), and all result files are available at https://github.com/ccgomezn/geometry-llm.

References

- An, J., Yang, L., Wang, M., Lu, L., An, W., & Yang, E. (2026). From Human Cognition to Neural Activations: Probing the Computational Primitives of Spatial Reasoning in LLMs. arXiv:2603.26323.
- Engels, J., Michaud, E. J., Liao, I., Gurnee, W., & Tegmark, M. (2024). Not All Language Model Features Are One-Dimensionally Linear. *ICLR 2025*. arXiv:2405.14860.
- Feng, J., Russell, S., & Steinhardt, J. (2024). Monitoring Latent World States in Language Models with Propositional Probes. arXiv:2406.19501.
- Geiger, A., Wu, Z., Potts, C., Icard, T., & Goodman, N. (2024). Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations. *CLeaR 2024*, PMLR 236, 160–187. arXiv:2303.02536.
- Gurnee, W., & Tegmark, M. (2023). Language Models Represent Space and Time. *ICLR 2024*. arXiv:2310.02207.
- Hewitt, J., & Liang, P. (2019). Designing and Interpreting Probes with Control Tasks. *EMNLP-IJCNLP*, 2733–2743. doi:10.18653/v1/D19-1275
- Kosslyn, S. M., Ball, T. M., & Reiser, B. J. (1978). Visual images preserve metric spatial information: Evidence from studies of image scanning. *JEP:HPP*, 4(1), 47–60. doi:10.1037/0096-1523.4.1.47
- Makelov, A., Lange, G., Geiger, A., & Nanda, N. (2024). Is This the Subspace You Are Looking for? An Interpretability Illusion for Subspace Activation Patching. *ICLR 2024*. arXiv:2311.17030.

- Pylyshyn, Z. W. (2002). Mental imagery: In search of a theory. *BBS*, 25(2), 157–182. doi:10.1017/S0140525X02000043
- Ravfogel, S., Elazar, Y., Gonen, H., Twiton, M., & Goldberg, Y. (2020). Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection. *ACL 2020*, 7237–7256. doi:10.18653/v1/2020.acl-main.647. arXiv:2004.07667.
- Tehenan, M., Bolivar Moya, C., Long, T., & Lin, G. (2025). Linear Spatial World Models Emerge in Large Language Models. arXiv:2506.02996.
- Vafa, K., Chen, J. Y., Rambachan, A., Kleinberg, J., & Mullainathan, S. (2024). Evaluating the World Model Implicit in a Generative Model. *NeurIPS 2024*. arXiv:2406.03689.